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INFORMATION VOLUME FRACTAL DIMENSION

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Abstract

There has been immense interest in uncertainty measurement because most real-world problems are accompanied by uncertain events. Therefore, Deng entropy has been proposed to measure the uncertainty in the probability theory and evidence theory. In this paper, we show that the uncertainty of the basic probability assignment (BPA) separated through the maximum Deng entropy separation rule (MDESR) is larger than the maximum Deng entropy of

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the original BPA. In addition, when the cardinality of the frame of discernment increases, the maximum information volume becomes larger and converges slower. The information volume fractal dimension is then proposed to describe the fractal property of uncertainty about the separated BPA distribution, which indicates the inherent physical meanings of Deng entropy from the perspective of statistics. This work can inspire further research on the fractal property of Deng entropy. Some experiments are applied to show the applicability of our proposed information volume fractal dimension.

Keywords: Information Volume; Deng Entropy; Fractal Property; Mass Function.

1. INTRODUCTION

In recent times, the way to express and deal with the uncertainty in the open world has become a topic of immense interest, and several theories have been proposed to measure the uncertainty, such as probability theory,[1,2] fuzzy set theory,[3] rough set theory,[4] Dempster–Shafer's evidence theory,[5–8] and Z numbers.[9] Due to the effectiveness of these approaches in the uncertain environment, they have been widely applied to numerous fields, including decision-making,[10–13] information fusion,[14–16] game theory,[17–19] artificial intelligence,[20–22] ecosystem,[23–25] risk analysis,[26–28] and complex network.[29–32] In information theory, Shannon entropy[33] was proposed to measure the information contained in the transmitted message, which describes the uncertainty and randomness of systems. For the uniform probability set, Shannon entropy will reach the maximum value which means the maximum uncertainty in the event, and it will gradually decrease when the probability becomes nonuniform. More generalized entropies were then developed to solve the real-world problems,[34–36] such as Boltzmann–Gibbs entropy,[37] Tsallis entropy,[38] and Rényi entropy.[39]

In order to measure the uncertainty of probability set in probability theory and BPA in evidence theory, Deng entropy[40] was proposed, which is a generalized form of Shannon entropy.[41] Deng entropy can degenerate to Shannon entropy when BPA degenerates to probability set. Compared with the traditional approaches, Deng entropy is more reasonable because it considers both discord and non-specificity. Kang *et al.* found that the uniform distribution cannot yield the maximum Deng entropy.[40] Deng entropy will reach the maximum value when BPA is distributed in a fractal structure, and the maximum information volume measured by Deng entropy is much larger than the maximum information volume of previous belief entropy functions measured by Shannon entropy. However, Deng *et al.* found that the maximum information volume of mass function can increase and gradually converge when BPA is separated based on the rule.[40] In addition, the maximum information volume will be much larger than the uncertainty measured by Deng entropy. This rule is further applied to the fuzzy membership function, including classic fuzzy sets and intuitionistic fuzzy sets.[42]

In this paper, we developed a new dimension to show how the maximum information volume changes according to the cardinality of the frame of discernment, which is called the information volume fractal dimension. When BPA is separated through the MDESR algorithm, the maximum information volume measured by Deng entropy will be larger than the maximum Deng entropy. In addition, the maximum information volume will increase and need more iterations to become convergence when the cardinality of the frame of discernment is larger. We also found that the maximum information volume and the Deng information scale follow the power-law relationship, which shows the fractal property. The information volume fractal dimension equals 0.6090 when the initial mass function can cause the maximum Deng entropy. This indicates that further works related to the fractal and scale-free property can be analyzed by Deng entropy. Some experiments are applied to show the applicability of our proposed information volume fractal dimension.

The rest of this paper is organized as follows. Section 2 introduces some concepts about Dempster–Shafer's evidence theory, Deng entropy, and fractal dimension. This proposed information volume fractal dimension is proposed in Sec. 3. Some numerical experiments are simulated to illustrate this proposed method in Sec. 4. The conclusions are discussed in Sec. 5.

2. PRELIMINARIES

In this section, several concepts about evidence theory and fractal theory are introduced to facilitate the understanding of this proposed model.

2.1. Dempster–Shafer's Evidence Theory

To deal with uncertainty in the open world, Dempster–Shafer's evidence theory[5,7] has been proposed and widely applied in real-world problems. The frame of discernment Θ is the set of collectively exhaustive and mutually exclusive events θ_i with N elements, which can be represented as

$$\Theta = \{\theta_1, \theta_2, \theta_3, \ldots, \theta_N\}. \tag{1}$$

In the power set of Θ, there are 2^N elements. Thus, the power set 2^Θ contains all possible subsets of Θ, which is shown as

$$\begin{aligned} 2^\Theta &= \{A_1, A_2, A_3, \ldots, A_{2^N}\} \\ &= \{\emptyset, \{\theta_1\}, \ldots, \{\theta_N\}, \{\theta_1, \theta_2\}, \ldots, \\ &\qquad \{\theta_1, \theta_N\}, \{\theta_1, \theta_2, \theta_3\}, \ldots, \Theta\}, \end{aligned} \tag{2}$$

where A_i is the focal element of the power set Θ. For the frame of discernment Θ, the mass function can map m from 2^Θ to $[0, 1]$,

$$m : 2^\Theta \to [0, 1] \tag{3}$$

which satisfies

$$\begin{aligned} \sum_{A \in 2^\Theta} m(A) = 1, \\ m(\emptyset) = 0, \end{aligned} \tag{4}$$

where $\emptyset$ is the empty set, and $m(A)$ shows the strength of the evidence to support event A.

2.2. Deng Entropy

In order to measure the uncertainty of systems, several different entropy have been proposed,[43] such as Shannon entropy,[33] Tsallis entropy,[38] and Rényi entropy.[39] However, the existing approaches cannot measure the uncertainty of BPA in evidence theory effectively. Therefore, Deng entropy,[40] a generalization of Shannon entropy, has been developed,

$$H(m) = -\sum_{A \in 2^\Theta} m(A) \log\left(\frac{m(A)}{2^{|A|} - 1}\right), \tag{5}$$

where $m(A)$ is the mass function defined in Θ, A represents the focal element of m, and $|A|$ is the cardinality of A. Observed from Eq. (5), the difference between Deng entropy and Shannon entropy is that the mass function $m(A)$ will be divided by the potential number of states of A, that is, $2^{|A|} - 1$. It can be easily rewritten via transformation, which is shown as

$$\begin{aligned} H(m) = &\sum_{A \in 2^\Theta} \log(2^{|A|} - 1) \\ &- \sum_{A \in 2^\Theta} m(A) \log m(A). \end{aligned} \tag{6}$$

Deng entropy will degenerate into Shannon entropy when all focal elements become singleton, that is, BPA degenerates to probability set. Due to its nonspecificity and discord, it has been used as an effective approach to measure the total uncertainty in the open world.

In order to measure the uncertainty more properly, the maximum Deng entropy has been studied.[40] The analytic solution of the maximum Deng entropy has been derived. They found that Deng entropy can reach its maximum value when the BPA is given as

$$m(A) = \frac{(2^{|A|} - 1)}{\sum_{A \in 2^\Theta} (2^{|A|} - 1)}. \tag{7}$$

Hence, the maximum Deng entropy is shown as

$$\bar{H}(m) = \log \sum_{A \in 2^\Theta} (2^{|A|} - 1). \tag{8}$$

Interestingly, they found that the maximum information volume obtained by Deng entropy $\bar{H}(m)$ is much bigger than the maximum information volume obtained by other measures.

2.3. Fractal Property

The fractal property was first proposed to measure the length of the coast of Britain.[44] The length of the coast is different by using different scales, and the length will be incorrect under inappropriate scale measurement. Therefore, Benoit B. Mandelbrot has found that objects have both integer dimension and fractal dimension. The fractal dimension d describes the complexity of objects and shows how the fractal pattern changes with different scales, which is shown as

$$d = \lim_{s \to 0} -\frac{\ln N(s)}{\ln s}, \tag{9}$$

where s and $N(s)$ are the scaled size and the number of self-similar pieces, respectively. An object

will show the fractal property when s and $N(s)$ obey the double logarithmic relationship.[45,46] It has been applied to address several practical problems, including robustness of networks,[47] vulnerability of networks,[48] and impacts of content of phosphorus slag.[49]

3. INFORMATION VOLUME FRACTAL DIMENSION

In order to obtain the information volume fractal dimension, the maximum information volume of the mass function needs to be obtained via an iteration process. The fractal property is then explored based on the BPA distribution and the maximum information volume.

3.1. Information Volume of Mass Function

Given a frame of discernment $\Theta = \{\theta_1, \theta_2, \theta_3, \ldots, \theta_N\}$ and corresponding power set 2^{Θ}, the mass function is shown as $m^0(A_j)$. The mass function $m^0(A_j)$ is then separated based on the maximum Deng entropy separation rule (MDESR)[40] when the cardinality of element is larger than 1 ($|A_j| > 1$). The proportion of BPA separated from $m^i(A_j)$ is given by

$$p(A_j^k) = \frac{(2^{|A_j^k|} - 1)}{\sum_{A_j^k \in 2^{A_j}} (2^{|A_j^k|} - 1)}. \tag{10}$$

Therefore, there will be $2^{|A_j|}$ new mass functions, and the kth mass function in the $i+1$th iteration separated from $m^i(A_j)$ will be $m^i(A_j) \times p(A_j^k)$. The framework of MDESR is shown in Algorithm 1. These new mass functions should satisfy the condition

$$\begin{aligned} &\sum_{k=1}^{2^{|A_j|}} m^{i+1}({A_j}^k) = m^i(A_j), \\ &m^{i+1}({A_j}^1) : m^{i+1}({A_j}^2) : \cdots : m^{i+1}({A_j}^{2^{|A_j|}}) \\ &\quad = p(A_j^1) : p(A_j^2) : \cdots : p(A_j^{2^{|A_j|}}). \end{aligned} \tag{11}$$

Based on the new mass function, the uncertainty $H_i(m^i)$ can be measured by Deng entropy, and it is shown as

$$H_i(m^i) = -\sum_{A_j \in 2^{\Theta}} m^i(A_j) \log\left(\frac{m^i(A_j)}{2^{|A_j|} - 1}\right). \tag{12}$$

Algorithm 1. Framework of maximum Deng entropy separation rule (MDESR).

Input: Mass function $m^i(A_j)$; cardinality $|A_j|$;
Output: New mass function $m^{i+1}(A_j)$;

1: The frame of discernment is A_j and corresponding power set is 2^{A_j};
2: In order to render the Deng entropy to reach the maximum value, $m^i(A_j)$ should be divided into $2^{|A_j|}$ parts and the proportion follows Eq. (10);
3: **for** $k = 1$ **to** $2^{|A_j|}$ **do**
4: $\quad m^{i+1}(A_j^k) = m^i(A_j) \times p(A_j^k)$;
5: **end for**
6: The mass function needs to satisfy the condition shown in Eq. (11);

When the difference of uncertainty between different iterations $\Delta = H_i(m^i) - H_{i-1}(m^{i-1})$ is smaller than the allowable error ε, the information volume of the mass function $H_M(m)$ can be obtained and shown as

$$H_M = -\sum_{F_i \in 2^{\Theta}} m(F_i) \log_2 \frac{m(F_i)}{2^{|F_i|} - 1}, \tag{13}$$

where $m(F_i)$ is the mass function of the last iteration. The framework is shown in Algorithm 2.

3.2. Information Volume Fractal Dimension

The information volume fractal dimension is proposed to describe the relationship between the maximum information volume $H_M(m)$ and the Deng information scale r,

$$H_M(m) \sim r^{-d_f}, \tag{14}$$

where d_f represents the information volume fractal dimension, and the Deng information scale r equals $\sum_{F_i \in 2^{\Theta}} (2^{|F_i|} - 1)$ in our model. The information volume fractal dimension d_f can be re-written as follows:

$$d_f = \lim_{|\Theta| \to \infty} \frac{H_M(m)}{\log_2 \sum_{F_i \in 2^{\Theta}} (2^{|F_i|} - 1)}, \tag{15}$$

where $H_M(m)$ is the information volume of the mass function obtained from Eq. (13), and $m(F_i)$ is the mass function of the last iteration that reaches the maximum information volume. The framework to obtain the information volume fractal dimension is shown in Algorithm 3.

Algorithm 2. Framework to obtain the maximum information volume.

Input: Initial mass function $m^0(A_j)$; Allowable error ε; Initial error $\Delta > \varepsilon$; Step $i = 1$;
Output: Maximum information volume of the mass function $H_M(m)$;

```
Calculate the uncertainty of mass function H_0(m) by Eq. (5);
while Δ > ε do
    for j = 1 to 2^N do
        if |A_j| > 1 then
            Separate the mass function of these elements to m^i(A_j) based on the maximum Deng entropy rule (Algorithm 1);
        else
            The mass function is unchanged m^i(A_j) = m^{i-1}(A_j);
        end if
    end for
    Obtain the uncertainty of the new mass function H_i(m^i) by Eq. (12);
    Calculate the error Δ = H_i(m^i) − H_{i-1}(m^{i-1});
    i = i + 1;
end while
```

Algorithm 3. Framework to obtain the information volume fractal dimension.

Input: Initial mass function $m(A_j)$; Frame of discernment Θ;
Output: Information volume fractal dimension d_f;

```
for Θ = 2 to Infinite ∞ do
    For a given Θ, the initial mass function should follow the distribution that can cause the maximum Deng entropy;
    Calculate the maximum information volume H_M(m) with the initial mass function separated by Algorithm 2;
    Obtain the Deng information scale r = Σ_{F_i ∈ 2^Θ} (2^{|F_i|} − 1);
    Record the maximum information volume H_M(m) and the Deng information scale r in each step;
end for
Calculate the information volume fractal dimension d_f based on Eq. (15);
```

4. NUMERICAL EXAMPLES

The distribution that can cause the maximum Deng entropy needs to be modified to achieve the maximum information volume. Here, the distribution when $|\Theta| = 2$ is used to show the maximum Deng entropy separation rule (MDESR). The initial mass function is shown as $m(X_0) = 1/5$, $m(Y_0) = 1/5$, and $m(U_0) = 3/5$, where $\{U_0\} = \{X_0, Y_0\}$. Since there is only one element in $\{X_0\}$ and $\{Y_0\}$, these two mass functions $m(X_0)$ and $m(Y_0)$ will not be separated. The mass function $m(U_0)$ will be separated into three parts with the proportion $(\frac{1}{5} : \frac{1}{5} : \frac{3}{5})$ because there are two elements. The new mass function after 1st iteration is shown as

$$\begin{cases} m(X_0) = \dfrac{1}{5}, \\ m(Y_0) = \dfrac{1}{5}, \\ m(X_1) = \dfrac{1}{5} \times \dfrac{3}{5} = \dfrac{3}{25}, \\ m(Y_1) = \dfrac{1}{5} \times \dfrac{3}{5} = \dfrac{3}{25}, \\ m(U_1) = \dfrac{3}{5} \times \dfrac{3}{5} = \dfrac{9}{25}, \end{cases} \tag{16}$$

where $m(X_1)$, $m(Y_1)$ and $m(U_1)$ are obtained from $m(U_0)$. The uncertainty of the new mass function can be measured by Deng entropy shown in Eq. (12),

$$\begin{aligned} H_1 = & -\frac{1}{5}\log_2\left(\frac{1}{5}\right) - \frac{1}{5}\log_2\left(\frac{1}{5}\right) \\ & - \frac{3}{25}\log_2\left(\frac{3}{25}\right) - \frac{3}{25}\log_2\left(\frac{3}{25}\right) \\ & - \frac{9}{25}\log_2\left(\frac{9/25}{3}\right) \\ = & \ 2.76411. \end{aligned} \tag{17}$$

It is easy to find that the uncertainty of the new BPA distribution is larger than the maximum Deng entropy $2.76411 > 2.32193$, which means that the uncertainty increases when the BPA is separated.

The mass function $m(U_1)$ needs further separation because there are two elements, and the other mass function will not be separated. The detailed process is shown in Fig. 1a which is called MDESR2. The iteration will stop until convergence, that is, the difference Δ is smaller than the allowable error ε. The distribution with $|\Theta| = 3$ is more complicated because there are more mass functions whose cardinality equal 2 and 3 at the same time, and MDESR3 is shown in Fig. 1b. The separation rule for the distribution whose cardinality is larger

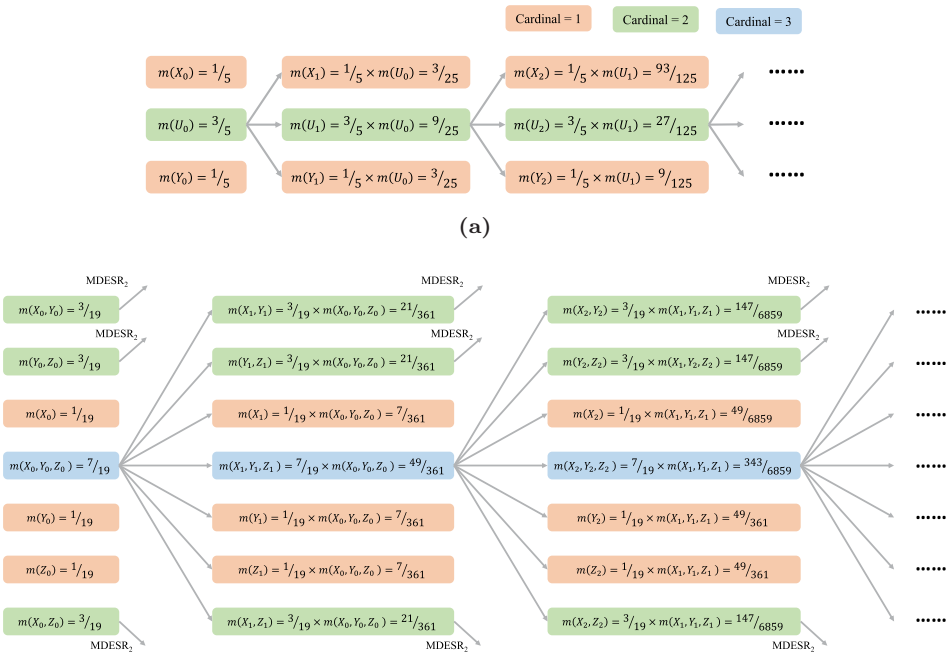

Fig. 1 The maximum Deng entropy separation rule. **(a)** The maximum Deng entropy separation rule when $|\Theta| = 2$ (MDESR2) and **(b)** The maximum Deng entropy separation rule when $|\Theta| = 3$ (MDESR3).

than 3 can be easily derived from MDESR2 and MDESR3 shown in Fig. 1, thus it will not be discussed in this paper.

The BPA distribution with different cardinalities $|\Theta|$ that can cause the maximum Deng entropy is

Table 1 The Maximum Information Volume $H_M(m)$ Under Different $|\Theta|$ in the ith Iteration.

ith	$\|\Theta\| = 2$	$\|\Theta\| = 3$	$\|\Theta\| = 4$	$\|\Theta\| = 5$	$\|\Theta\| = 6$
1	2.3219	4.2479	6.0224	7.7211	9.3772
2	2.7641	5.1278	7.3352	9.4623	11.5421
3	3.0294	5.6614	8.1396	10.5395	12.8936
4	3.1886	5.9836	8.6286	11.1984	13.7253
5	3.2841	6.1777	8.9243	11.5986	14.2325
6	3.3414	6.2945	9.1026	11.8404	14.5399
7	3.3758	6.3647	9.2099	11.9863	14.7254
8	3.3964	6.4068	9.2745	12.0740	14.8372
9	3.4088	6.4321	9.3132	12.1267	14.9044
10	3.4162	6.4473	9.3365	12.1584	14.9448
11	3.4207	6.4564	9.3504	12.1774	14.9691
12	3.4234	6.4619	9.3588	12.1888	14.9836
13	3.4250	6.4652	9.3639	12.1957	14.9924
14	3.4259	6.4671	9.3669	12.1998	14.9976
15	—	6.4683	9.3687	12.2023	15.0008
16	—	6.4690	9.3698	12.2037	15.0027
17	—	—	9.3704	12.2046	15.0038
18	—	—	—	—	15.0045

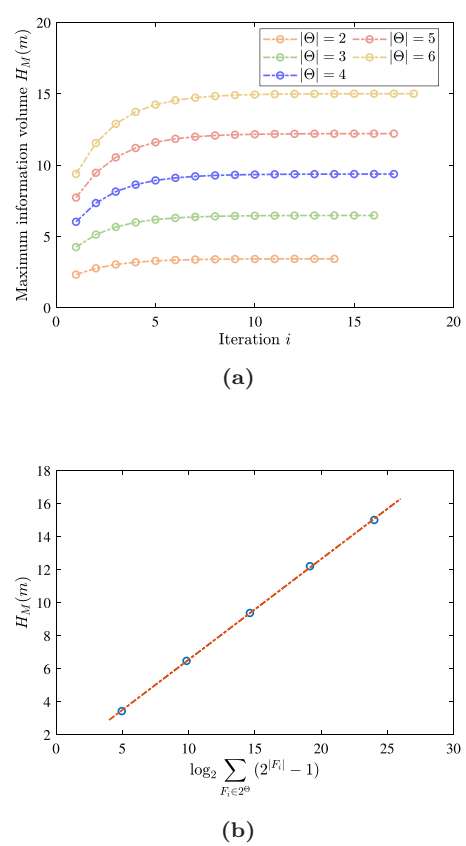

Fig. 2 The maximum information volume under different cases. **(a)** $H_M(m)$ with different $|\Theta|$ and **(b)** The double logarithmic scale fitting curves.

Table 2 $H_M(m)$ **and** $\log_2 \sum_{F_i \in \Theta} (2^{|F_i|} - 1)$ **in Different Cardinalities.**

$\lvert\Theta\rvert$	2	3	4	5	6
$\log_2 \sum_{F_i \in \Theta} (2^{\lvert F_i\rvert} - 1)$	4.9542	9.8439	14.6294	19.1618	24.0151
$H_M(m)$	3.4259	6.4690	9.3704	12.2046	15.0045
ith step	14	16	17	17	18

used in this paper as the initial BPA distribution. Therefore, the initial BPA distribution follows the condition $m(F_i) = \frac{2^{|F_i|-1}}{\sum_i 2^{|F_i|-1}}$. These mass functions whose cardinality are larger than 1 will be separated to new mass function based on the MDESR ($\varepsilon = 0.001$), and the results are shown in Table 1. MDESR2 and MDESR3 when $|\Theta| = 2$ and $|\Theta| = 3$ are discussed in Fig. 1, and MDESR with more cardinality can be easily derived. From Table 1 and Fig. 2a, when $|\Theta|$ is smaller, there are fewer iterations to obtain the maximum information volume (H_i converges faster), and the maximum information volume $H_M(m)$ is smaller. With different $|\Theta|$, the Deng information scale $\log_2 \sum_{F_i \in \Theta} (2^{|F_i|} - 1)$, maximum information volume $H_M(m)$, and the number of iterations are shown in Table 2. Therefore, the information volume fractal dimension d_f can be obtained by the double logarithmic fitting from Eq. (14), and the power law relationship is shown in Fig. 2b, which shows the scale-free and fractal property. The information volume fractal dimension d_f is 0.6090 when the initial mass function reaches the maximum Deng entropy.

5. CONCLUSION

Deng entropy has been proposed to measure the uncertainty of the probability set in probability theory and BPA in evidence theory. However, the uncertainty of the new BPA distribution separated through MDESR is larger than the maximum Deng entropy of the original BPA distribution. We show that there are fewer iterations to obtain the maximum information volume, and the maximum information volume is smaller when the cardinality $|\Theta|$ is smaller. In addition, we proposed a novel dimension that is called the information volume fractal dimension to describe the change of uncertainty when there are different cardinalities of the frame of discernment. Interestingly, the separation of BPA obtained by MDESR shows the fractal property. This indicates further works related to the fractal and scale-free property can be analyzed by Deng entropy, and the results of this paper can inspire further research. Deng entropy can be then applied to show the fractal property of objects and networks, thereby discovering more information in the open world. In addition, other kinds of dimensions that can show the fractal property and discover more characteristics in the uncertainty measurement can be developed.

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